

Lung cancer prediction using machine learning

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Abstract. One of the most deadly diseases in the world today is lung cancer and increasing survival rates require early identification. Nevertheless, early diagnosis is frequently a challenge for conventional diagnostic techniques which causes therapy delays and subpar results. In this research work, machine learning models are used to evaluate the likelihood of lung cancer based on answers to a customized questionnaire in order to address this difficulty. Based on the examination of significant risk factors and symptoms, the proposed work produces a probability score that represents the possibility of lung cancer. People who have a high probability score are encouraged to get medical advice right away. This method uses cutting-edge algorithms to deliver a preliminary diagnostic that is quick, easy to get, and non-invasive - potentially supporting early detection and intervention. Our findings highlight the potential of machine learning to revolutionize early lung cancer detection, offering a valuable tool in the fight against this deadly disease.

Keywords: Lung cancer, early detection, machine learning, predictive modeling, risk assessment, medical diagnosis, questionnaire analysis, non-invasive diagnosis, healthcare technology

1 Introduction

Lung cancer is one of the leading causes of cancer-related deaths worldwide, accounting for nearly 1.8 million deaths annually [1]. The global burden of lung cancer is growing due to increasing exposure to risk factors such as smoking, air pollution, and occupational hazards. In India, lung cancer accounts for approximately 5.9% of all cancer cases and 8.5% of cancer-related deaths, making it a significant public health concern [2]. The trend of lung cancer incidences as per age and gender is presented in Fig. 1. Early detection of lung cancer improves treatment outcomes, yet the diagnosis often occurs at advanced stages due to the lack of effective screening mechanisms. Machine Learning (ML) and Artificial Intelligence (AI) offer promising approaches to improve the detection, diagnosis, and prediction of lung cancer [3]. Machine learning techniques can analyze complex medical data, identify patterns, and make predictions that aid clinicians in diagnosing diseases with higher accuracy [4]. The use of AI in healthcare is expanding rapidly, especially in areas such as imaging, genomics, and personalized medicine. Machine learning algorithms have been applied to predict cancer progression, survival rates, and response to treatments [5].

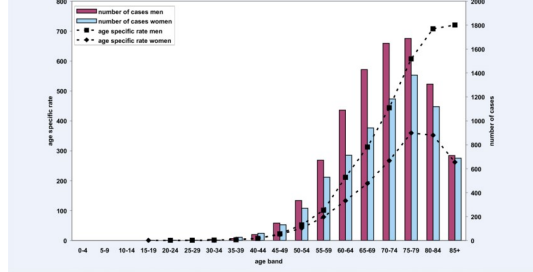


Fig. 1. Lung Cancer according to age and gender

Many studies have focused on leveraging ML models such as Support Vector Machines (SVM), Random Forest, Decision Trees, and K-Nearest Neighbors (KNN) to enhance the accuracy of lung cancer prediction [6]. Lung cancer remains a major cause of mortality, primarily because it is often diagnosed at later stages when treatment is less effective. Studies have shown that early detection significantly improves survival rates [7]. Non-invasive methods for screening, such as questionnaires and health surveys, are being explored to facilitate early detection [8]. Machine learning has been increasingly utilized in healthcare for disease diagnosis and prediction. Questionnaire-based approaches have also been utilized to predict lung cancer risk. Wu et al. [9] conducted a study using a questionnaire that collected information on smoking habits, exposure to environmental pollutants, and family history of cancer. In this context, questionnaires serve as a valuable tool for gathering large-scale, population-level data for ML applications [10].

In India, the integration of AI and ML into the healthcare sector is still in its nascent stage, but efforts are underway to develop AI-driven applications that can aid in cancer diagnosis and management. The Indian healthcare system is gradually adopting AI for automating diagnostic procedures and optimizing treatment plans [11].

This paper focuses on the application of various machine learning models to predict lung cancer using a questionnaire-based dataset. Questionnaires remain one of the simplest, cost-effective, and non-invasive methods of collecting health-related data. In this study, demographic information, health habits, and symptoms are used as input features for various machine learning models, which are trained to predict lung cancer. The accuracy and performance of models such as SVM, Random Forest, Decision Trees, and KNN are compared, and the impact of feature engineering and data preprocessing techniques on prediction accuracy is explored.

3 Methodology

The proposed system involves data collection, data preprocessing, and training various machine learning approaches for lung cancer prediction. An openly accessible source specializing in lung cancer diagnostics provided the dataset used in this investigation. It has a number of aspects that outline the habits, health issues, and demographics of

the patients. Every record has a binary classification, meaning it can either have lung cancer or not. The variables used as inputs are patient's details (Age, Gender), addictions (Smoking, Yellow Fingers, and Alcohol Consumption), mental health (Anxiety, Peer Pressure), and disease symptoms (Fatigue, Chronic Disease, Allergy, Wheezing, Coughing, Shortness of Breath, Swallowing Difficulty, and Chest Pain) as shown in Fig. 2.

	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFFICULTY	CHEST_PAIN	LUNG_CANCER
0	M	69	1	2	2	1	1	2	1	2	2	2	2	2	2	YES
1	M	74	2	1	1	1	2	2	2	1	1	1	2	2	2	YES
2	F	59	1	1	1	2	1	2	1	2	1	2	2	1	2	NO
3	M	63	2	2	2	1	1	1	1	1	2	1	1	2	2	NO
4	F	63	1	2	1	1	1	1	1	2	1	2	2	1	1	NO
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
304	F	56	1	1	1	2	2	2	1	1	2	2	2	2	1	YES
305	M	70	2	1	1	1	1	2	2	2	2	2	2	1	2	YES

Fig. 2. Few Samples of Dataset

Data preprocessing. It involves handling missing and duplicate data to ensure the dataset's integrity. Since ML algorithms work better with numerical data, categorical variables were label-encoded such as Gender was encoded as 0 (Male) and 1 (Female)) and Lung Cancer was encoded as 0 (No) and 1 (Yes). Other binary categorical variables were converted in numerical format as (0 or 1), so no additional transformation was required. Imbalanced data is handled to make a class balance. The input feature like age has a large numerical range which is standardized to ensure that all features contribute equally to the model's learning process. A heat map of feature correlations was generated to understand the relationships between different variables. This visualization helps to identify multicollinearity or strong relationships between features, which can be managed during feature selection.

Automated Learning Systems. The performance of various machine learning algorithms for detection of lung cancer using these parameters is evaluated in this study. Seven ML algorithms used were in this study i.e. Support Vector Machine (SVM), Logistic Regression (LogReg), Decision Tree Classifier (DT), Random Forest Classifier (RF), K-Nearest Neighbours (KNN), Naive Bayes (NB), XGBoost Classifier (XGBoost). The preprocessed dataset was used to train each of these models, with the hyperparameters left at default for ease of comparison. Using train test split, the dataset was divided into training (80%) and testing (20%) sets.

4 Results and Discussion

The lung cancer dataset was used to predict whether a patient has lung cancer based on demographic, behavioral, and health-related features. A total of 15 features were used to train several machine learning models. Initially, the models are trained without data preprocessing techniques. However, the models' performances are found to be suboptimal. To improve performance, following data preprocessing techniques are implemented. Feature Scaling and Normalization is used to ensure that all features contribute equally to the machine learning model. Visualization of Standardized Data

helped to identify certain features which are outliers. These outlier features are then treated using the Interquartile Range (IQR) method. Lastly, we conducted a correlation analysis to identify multicollinearity among the input features. Highly correlated features can introduce redundancy and lead to overfitting in models. The heatmap in Fig. 4 shows the correlation between the features.

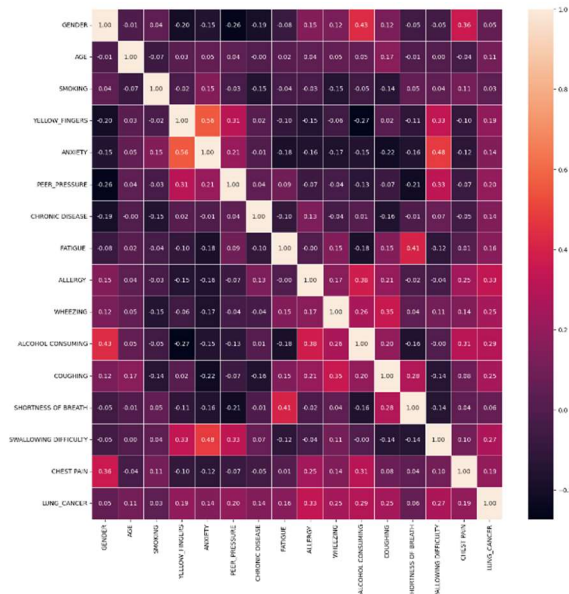


Fig. 4. Heatmap for correlation between features

Several features such as SMOKING and YELLOW FINGERS showed a strong correlation, indicating that one of these could be dropped to simplify the model.

4.1 Impact of Data Preprocessing

The preprocessing techniques significantly improved the models' performances. Table 1 shows the performance comparison of the machine learning algorithms before and after preprocessing.

Table 1. Impact of pre-processing and normalization of data.

Model	Accuracy (%) Before	Accuracy (%) After
RF	85.10	94.96
DT	83.25	93.28
SVM	79.45	89.92
KNN	76.35	89.92
NB	78.15	89.92
XGBoost	81.25	89.92
LogReg	75.12	82.71

As illustrated, Random Forest showed the highest improvement, achieving an accuracy of 94.96% after preprocessing, compared to 85.10% before preprocessing. Decision Tree also benefited significantly, with accuracy rising from 83.25% to 93.28%.

4.2 Model Performance Comparison

After applying feature scaling, outlier treatment, and correlation analysis, various machine learning models were trained on the dataset. Table 2 summarizes the models' performance, evaluating them using Accuracy, Precision, Recall, and F1 Score.

Table 2. Performance evaluation of machine learning algorithms for lung cancer prediction.

Model	Accuracy (%)	Precision	Recall	F1 Score
RF	94.96	1.000	0.898	0.946
DT	93.28	0.932	0.932	0.932
SVM	89.92	0.943	0.847	0.893
KNN	59.60	0.596	0.800	0.603
NB	89.92	1.000	0.797	0.887
XGBoost	89.92	1.000	0.797	0.887
LogReg	85.71	0.862	0.848	0.855

Random Forest emerged as the best-performing model, with an accuracy of 94.96%, precision of 1.00, and F1 score of 0.946. This model benefits from its ensemble nature, which combines multiple decision trees to provide a robust prediction mechanism. Decision Tree and SVM performed strongly, with accuracies of 93.28% and 89.92%, respectively. Decision Tree's interpretability and ability to handle complex interactions between features contributed to its success in this context. K-Nearest Neighbors (KNN), Naive Bayes, and XGBoost all showed similar performance, achieving an accuracy of 89.92%. Despite KNN's simplicity, it performed well after feature scaling. However, KNN and Naive Bayes models might not generalize well to large datasets due to their reliance on distance metrics and probabilistic assumptions, respectively.

Logistic Regression lagged behind other models, achieving 85.71% accuracy, suggesting that linear boundaries may not be sufficient to separate lung cancer patients from non-patients based on the selected features. Random Forest emerged as the best-performing model, with an accuracy of 94.96%, precision of 1.00, and F1 score of 0.946. This model benefits from its ensemble nature, which combines multiple decision trees to provide a robust prediction mechanism. Decision Tree and SVM performed strongly, with accuracies of 93.28% and 89.92%, respectively. Decision Tree's interpretability and ability to handle complex interactions between features contributed to its success in this context.

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assumptions, respectively. Logistic Regression lagged behind other models, achieving 85.71% accuracy, suggesting that linear boundaries may not be sufficient to separate lung cancer patients from non-patients based on the selected features.

4 Conclusion

In this research, we investigated the forecasting of lung cancer with different machine learning models. A grand total of 15 characteristics were examined, encompassing demographic, behavioral, and health-related aspects. Feature scaling, outlier handling, and correlation examination were employed to improve the performance of the model. By applying these preprocessing methods, we alleviated the impact of outliers and minimized feature duplication, resulting in notable enhancements in the performance of the models. Random Forest and Decision Tree proved to be more effective than other algorithms in the realm of machine learning models. Random Forest obtained a 94.96% accuracy, whereas Decision Tree achieved 93.28%. These models stand out because they can detect intricate patterns in the data and manage linear and non-linear relationships effectively. Even though it is simple, K-Nearest Neighbors (KNN) still did well, using nearby patterns to reach an accuracy of 89.92%. This study shows how feature engineering and preprocessing techniques can enhance lung cancer prediction models. Nevertheless, the research was constrained by the dataset's size, potentially impacting the models' applicability.

References

- [1] Siegel RL, Miller KD, Fuchs HE, Jemal A (2022) Cancer statistics, 2022. *CA: A Cancer Journal for Clinicians* 72(1):7–33
- [2] Indian Council of Medical Research (2020) Cancer Statistics in India, 2020. ICMR, India
- [3] Kaur S, Gupta N (2021) Ai and machine learning in lung cancer diagnosis: A review. *Computational and Structural Biotechnology Journal* 19:2326–2341
- [4] Esteva A, Kuprel B, Novoa RA, et al (2017) Dermatologist-level classification of skin cancer with deep neural networks. *Nature* 542(7639):115–118
- [5] Hosny M, Parmar C, Quackenbush J, Schwartz LH, Aerts HJ (2018) Artificial intelligence in radiology. *Nature Reviews Cancer* 18(8):500–510
- [6] LeCun Y, Bengio Y, Hinton G (2015) Deep learning. *Nature* 521(7553):436–444
- [7] Patil PK, Syed MA (2020) Using ai for lung cancer screening in the indian healthcare system. *Journal of Cancer*
- [8] Sharma S, Gupta P (2020) Non-invasive methods for early detection of lung cancer. *BMC Pulmonary Medicine* 20(1):245
- [9] Wu J, Adams M (2022) Questionnaire-based lung cancer risk prediction: Machine learning models and feature selection. *BMC Cancer* 22(1):523
- [10] Smith A, Johnson M (2020) Questionnaire approaches in cancer risk assessment: Machine learning insights. *European Journal of Cancer Care* 29(2):e13,248
- [11] Dubey A, Jain R (2020) Role of ai in improving lung cancer treatment in india. *Indian Journal of Cancer Research* 46(4):652–660 Lopez-Campos D, Fernandez-Villar T, Izquierdo-Alonso P (2021) Global trends in lung cancer: 2021 report. *European Respiratory Journal* 58(1):210–227